# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years building **Java, Spring, REST, SQL, and distributed applications**, applying object-oriented design, multithreading, reliability engineering, performance optimization, and AI-assisted development to scalable customer-facing systems.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Advanced NLP, Analysis of Algorithms

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *Los Angeles, CA*

- Architected asynchronous FastAPI services and Azure SQL workflows, processing 1K+ concurrent jobs at 3x throughput through **object-oriented design, state transitions, fault isolation, monitoring, multithreading, and production debugging**.
- Optimized data services for 500+ users, reducing dashboard latency 71% through **normalized schemas, indexed queries, caching, concurrency controls, requirements analysis, automated testing, and scalable production deployment decisions**.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, CA*

- Delivered React and Flask workflows supporting 150+ instructors and 10K+ daily requests through **RESTful services, code reviews, validation, automated testing, production troubleshooting, and collaborative cross-functional solution design**.
- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, reducing turnaround 60% while improving reliability through **parallel execution, structured logging, exception handling, and failure recovery**.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* *India*

- Designed Java APIs and distributed data pipelines using object-oriented patterns, AWS Glue, S3, and caching, improving response times 30% while reducing technical debt 20% across **business-critical microservices**.
- Automated 20+ unit and integration suites across Windows and Linux using PyTest, Mockito, Jenkins, and virtual machines, strengthening **CI/CD validation, regression detection, design reviews, and dependable releases**.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern - (VibrantChem)* *India*

- Built React and Node.js microservices for billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, authentication, SQL-backed transactional workflows, reusable components, and resilient service integrations**.
- Automated Docker and Jenkins deployments, reducing release time 50% while using Sentry, Azure Monitor, and Kibana for **application health monitoring, root-cause debugging, performance analysis, and reliability improvement**.

**LST Capitals** **November 2022 – January 2023**
*Web Developer Intern* *India*

- Translated client requirements into technical designs for 12+ payment and booking integrations, mapping **API contracts, authorization flows, data synchronization, failure scenarios, and scalable implementation choices** across stakeholders.
- Built Flask and Firebase services supporting 50K+ payment-status records with sub-200-millisecond responses, delivering **integration testing, performance validation, bug fixes, technical documentation, and responsive production support** through release.

## Skills

**Languages & Frameworks:** Java, Core Java, J2EE, Spring, Spring Boot, Python, TypeScript, JavaScript, Node.js, Flask, FastAPI, React
**Distributed Systems:** Microservices, REST APIs, HTTP, Kafka, Event Streaming, Real-Time Processing, Multithreading, Concurrency, Fault Tolerance, High Availability, Scalability
**Data:** SQL, PostgreSQL, Azure SQL, MySQL, NoSQL, MongoDB, Redis, Data Modeling, Transactions, Indexing, Caching, Database Synchronization
**Cloud & Delivery:** AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Monitoring, Logging, Release Management
**Engineering:** Object-Oriented Design, Design Patterns, Data Structures, Algorithms, SDLC, Unit Testing, Integration Testing, Debugging, Troubleshooting, Performance Optimization, Code Reviews, Design Reviews, AI/GenAI Tools

## Projects

**StreamPay Processing Engine** | *Java, Spring Boot, Kafka, PostgreSQL, Redis, Docker*

- Designed a **real-time payment processing system** with idempotent REST endpoints, Kafka event streams, transactional persistence, retry isolation, reconciliation, and horizontally scalable workers for reliable authorization, settlement, and payout workflows.

**Payment Reliability Lab** | *Java, Spring, MongoDB, Kubernetes, OpenTelemetry, GenAI*

- Built a **microservice reliability and debugging platform** combining distributed traces, failure injection, availability metrics, load tests, SQL/NoSQL comparisons, and AI-assisted incident summaries with verifiable evidence and remediation guidance.

**Chat2SQL** | *Java, TypeScript, SQL, Docker, Maven, Node.js* | Code

- Built a cross-platform **natural-language SQL workspace** supporting 10+ databases, schema synchronization, query validation, and Docker-based environments, enabling users to explore complex relational data through reproducible, database-aware workflows.